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Experiment: 3

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3. Probability Distributions and Bayes' Theorem

Scenario

A hospital studies patients' waiting times, which approximately follow a normal distribution. Generate synthetic waiting-time samples, calculate probabilities for specified intervals, and visualize the distribution

Algorithm

1. Start the program.
2. Import NumPy, SciPy, and Matplotlib libraries.
3. Set the average waiting time, standard deviation, and number of patients.
4. Generate random waiting times for the patients using a normal distribution.
5. Calculate the probability of waiting less than 20 minutes.
6. Calculate the probability of waiting between 20 and 40 minutes.
7. Calculate the probability of waiting more than 40 minutes.
8. Display the waiting-time details and probabilities in a summary table.
9. Draw a histogram and normal distribution curve to show the waiting times.
10. Draw a bar chart to compare the probabilities.
11. Display the plots with suitable titles and labels.
12. Stop the program.

Source code

```
import numpy as np

import pandas as pd

import matplotlib.pyplot as plt

from scipy.stats import norm


# 1. Generate synthetic hospital waiting-time samples
```

```
np.random.seed(42)

mean = 30       # Mean waiting time in minutes
std_dev = 8     # Standard deviation in minutes
n = 1000        # Number of patients

waiting_times = np.random.normal(mean, std_dev, n)

# Waiting times cannot be negative
waiting_times = np.clip(waiting_times, 0, None)

# 2. Calculate probabilities using normal distribution
p_20_40 = norm.cdf(40, loc=mean, scale=std_dev) - norm.cdf(20, loc=mean,
scale=std_dev)
p_below_20 = norm.cdf(20, loc=mean, scale=std_dev)
p_above_40 = 1 - norm.cdf(40, loc=mean, scale=std_dev)

# 3. Create summary table
summary = pd.DataFrame({
   "Measure": [
      "Number of Patients",
      "Mean Waiting Time (min)",
      "Standard Deviation (min)",
      "Minimum Waiting Time (min)",
      "Maximum Waiting Time (min)",
      "Probability: Below 20 min",
      "Probability: 20 to 40 min",
      "Probability: Above 40 min"
   ],
```

```
    "Value": [
        n,
        np.mean(waiting_times),
        np.std(waiting_times, ddof=1),
        np.min(waiting_times),
        np.max(waiting_times),
        p_below_20,
        p_20_40,
        p_above_40
    ]
})

print("HOSPITAL WAITING TIME SUMMARY")
print(summary.round(3).to_string(index=False))

# 4. Display probability results as percentages
print("\nPROBABILITY RESULTS")
print(f"Waiting time below 20 minutes: {p_below_20 * 100:.2f}%")
print(f"Waiting time between 20 and 40 minutes: {p_20_40 * 100:.2f}%")
print(f"Waiting time above 40 minutes: {p_above_40 * 100:.2f}%")

# 5. Plot histogram and normal distribution curve
plt.figure(figsize=(9, 5))

plt.hist(
    waiting_times,
    bins=30,
    density=True,
    alpha=0.6,
```

```
    edgecolor="black",
    label="Simulated waiting times"
)

x = np.linspace(0, 65, 500)
y = norm.pdf(x, loc=mean, scale=std_dev)

plt.plot(x, y, linewidth=2, label="Theoretical normal distribution")

plt.axvline(20, linestyle="--", label="20 minutes")
plt.axvline(40, linestyle="--", label="40 minutes")

plt.title("Hospital Waiting Times - Normal Distribution")
plt.xlabel("Waiting Time (minutes)")
plt.ylabel("Probability Density")
plt.legend()
plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()

# 6. Plot probability comparison
labels = ["Below 20 min", "20–40 min", "Above 40 min"]
probabilities = [p_below_20, p_20_40, p_above_40]

plt.figure(figsize=(8, 5))
bars = plt.bar(labels, np.array(probabilities) * 100)

plt.title("Probability of Hospital Waiting Time Intervals")
plt.xlabel("Waiting Time Interval")
```

```
plt.ylabel("Probability (%)")
plt.ylim(0, 100)

for bar, probability in zip(bars, probabilities):
    plt.text(
        bar.get_x() + bar.get_width() / 2,
        bar.get_height() + 1,
        f"{probability * 100:.2f}%",
        ha="center"
    )

plt.tight_layout()
plt.show()
```

Output

```
HOSPITAL WAITING TIME SUMMARY
                   Measure    Value
         Number of Patients 1000.000
    Mean Waiting Time (min)   30.155
  Standard Deviation (min)    7.834
Minimum Waiting Time (min)    4.070
Maximum Waiting Time (min)   60.822
 Probability: Below 20 min    0.106
 Probability: 20 to 40 min    0.789
 Probability: Above 40 min    0.106

PROBABILITY RESULTS
Waiting time below 20 minutes: 10.56%
Waiting time between 20 and 40 minutes: 78.87%
Waiting time above 40 minutes: 10.56%
```

Waiting time above 40 minutes: 10.56%



